**Task 2 output and reflection**

**2a Results for rule-based stemming**

Apply on test words

English stemming:

playing -> play

played -> play

walking -> walk

sing -> s

quickly -> quick

cats -> cat

dogs -> dog

his -> hi

news -> new

boxes -> box

byes -> by

studies -> studi

misunderstand -> understand

disagree -> agree

disturb -> turb

mister -> ter

unhappy -> happy

until -> til

German stemming:

unglücklich -> glücklich

verkaufen -> kauf

gesagt -> sagt

zahlen -> zahl

Hunde -> Hund

Heiterkeit -> Heiter

Autos -> Auto

kannst -> kann

unklar -> klar

Gesicht -> sicht

gestern -> stern

Ungeheuer -> geheuer

Universität -> iversität

Keks -> Kek

Fenster -> Fenster

Verb -> b

verbinden -> bind

Ast -> A

Keule -> Keul

verkaufst -> kauf

Apply on sentences

English sentences:

The children are playing with their dogs.

['the', 'children', 'are', 'play', 'with', 'their', 'dog']

She quickly opened the boxes.

['she', 'quick', 'open', 'the', 'box']

The unhappy student disagreed with the teacher.

['the', 'happy', 'student', 'agre', 'with', 'the', 'teacher'] -> opposite meaning now

They are walking and singing in the garden.

['they', 'are', 'walk', 'and', 'sing', 'in', 'the', 'garden']

He misunderstood the question and replayed the video.

['he', 'understood', 'the', 'question', 'and', 'replay', 'the', 'video']

German sentences:

Die Kinder spielen mit ihren Hunden.

['di', 'kinder', 'spiel', 'mit', 'ihr', 'hund']

Der unglückliche Schüler verkauft seine Bücher.

['der', 'glücklich', 'schüler', 'kauft', 'sein', 'bücher'] -> now opposite Meaning

Sie bezahlten gestern die Einkäufe.

['si', 'bezahlt', 'stern', 'di', 'einkäuf']

Die Verkäufer arbeiten meistens sehr schnell.

['di', 'käufer', 'arbeit', 'meisten', 'sehr', 'schnell']

Das Geschenk wurde gestern ausgepackt und geöffnet.

['da', 'schenk', 'wurd', 'stern', 'ausgepackt', 'd', 'öffnet']

Task 2 b) Byte Pair Encoding (BPE)
Task 2.2 Select two different corpora

**English Corpus**

{"test": 1, "words": 1, "English": 1, "playing": 1, "played": 1, "cats": 1, "boxes": 1, "byes": 1, "dogs": 1, "his": 1, "walking": 1, "studies": 1, "sing": 1, "news": 1, "misunderstand": 1, "disagree": 1, "disturb": 1, "mister": 1, "unhappy": 1, "until": 1}

**Merged corpora after 16 Turns**

{'test': 1, 'words': 1, 'English': 1, 'playing': 1, 'played': 1, 'cats': 1, 'boxes': 1, 'byes': 1, 'dogs': 1, 'his': 1, 'walking': 1, 'studies': 1, 'sing': 1, 'news': 1, 'misunderstand': 1, 'disagree': 1, 'disturb': 1, 'mister': 1, 'unhappy': 1, 'until': 1}

After 16 iterations, only pairs of tokens, each occurring once in the training vocabulary, remained. Further iterations generated individual words again, until the entire corpus was once more represented as individual words in the token list. With this method, it appears to be important to define a suitable termination condition for token generation.

Final token list after 16 iterations

['t', 'es', 'w', 'o', 'r', 'd', 's', 'Engl', 'is', 'h', 'play', 'ing', 'e', 'c', 'a', 'b', 'x', 'y', 'g', 'l', 'k', 'st', 'u', 'i', 'n', 'mis', 'un', 'er', 'dis', 'p']

**Application of the BPE model to an unknown text with language match (English – English)**

The children are playing with their dogs.
['T h e', 'c h i l d r e n', 'a r e', 'play ing', 'w i t h', 't h e i r', 'd o g s '.]

She quickly opened the boxes.
['S h e', 'q u i c k l y', 'o p e n e d', 't h e', 'b o x es .']

The unhappy student disagreed with the teacher.
['T h e', 'un h a p p y', 'st u d e n t', 'dis a g r e e d', 'w i t h', 't h e', 't e a c h er .']

They are walking and singing in the garden.
['T h e y', 'a r e', 'w a l k ing', 'a n d', 's ing ing', 'i n', 't h e', 'g a r d e n .']

He misunderstood the question and replayed the video.
, ['H e', 'mis un d er st o o d', 't h e', 'q u es t i o n', 'a nd', 'r e play e d', 't h e', 'v i d e o .']

**Language Mitsmatch English training – German unknown text:**

segmented: ['D i e', 'K i n d er', 's p i e l e n', 'm i t', 'i h r e n', 'H un d e n .', 'D er', 'un g l ü c k l i c h e', 'S c h ü l er', 'v er k a u f t', 's e i n e', 'B ü c h er .', 'S i e', 'b e z a h l t e n', 'g es t er n', 'd i e', 'E i n k ä u f e .', 'D i e', 'V er k ä u f er', 'a r b e i t e n', 'm e is t e n s', 's e h r', 's c h n e l l .', 'D a s', 'G es c h e n k', 'w u r d e', 'g es t er n', 'a u s g e p a c k t', 'un d', 'g e ö f f n e t .']

**German Corpus**

{ "u n g l ü c k l i c h": 1, "v e r k a u f e n": 1, "g e s a g t": 1, "z a h l e n": 1, "H u n d e": 1, "H e i t e r k e i t": 1, "A u t o s": 1, "k a n n s t": 1, "u n k l a r": 1, "G e s i c h t": 1, "g e s t e r n": 1, "U n g e h e u e r": 1, "U n i v e r s i t ä t": 1, "K e k s": 1, "F e n s t e r": 1, "V e r b": 1, "v e r b i n d e n": 1, "A s t": 1, "K e u l e": 1, "v e r k a u f s t": 1}

**Merged corpus after 18 Turns**

{'un g l ü c kl ich': 1, 'ver kauf en': 1, 'ge s a g t': 1, 'z a h l en': 1, 'H un d e': 1, 'H eit er k eit': 1, 'A u t o s': 1, 'ka n n st': 1, 'un kl a r': 1, 'G e s ich t': 1, 'ge ster n': 1, 'Un ge h e u er': 1, 'Un i ver s it ä t': 1, 'Ke k s': 1, 'F en ster': 1, 'V er b': 1, 'ver b i n d en': 1, 'A st': 1, 'Ke u l e': 1, 'ver kauf st': 1}

In the German corpus, after 18 passes, only token pairs each occurring once remained.

**Final token list after 18 turns**

['un', 'g', 'l', 'ü', 'c', 'kl', 'ich', 'ver', 'kauf', 'en', 'ge', 's', 'a', 't', 'z', 'h', 'H', 'd', 'e', 'eit', 'er', 'k', 'A', 'u', 'o', 'ka', 'n', 'st', 'r', 'G', 'ster', 'Un', 'i', 'it', 'ä', 'Ke', 'F', 'V', 'b']

**Application of the BPE model to an unknown text with language match (German – German)**

Die Kinder spielen mit ihren Hunden.
['D i e', 'K i n d er', 's p i e l en', 'm it', 'i h r en', 'H un d en .']
Der unglückliche Schüler verkauft seine Bücher.
['D er', 'un g l ü c kl ich e', 'S c h ü l er', 'ver kauf t', 's e i n e', 'B ü c h er .']
Sie bezahlten gestern die Einkäufe.
['S i e', 'b e z a h l t en', 'ge ster n', 'd i e', 'E i n k ä u f e .]
Die Verkäufer arbeiten meistens sehr schnell.
['D i e', 'V er k ä u f er', 'a r b eit en', 'm e i st en s','s e h r', 's c h n e l l .']
Das Geschenk wurde gestern ausgepackt und geöffnet.
['D a s', 'G e s c h en k', 'w u r d e', 'ge ster n', 'a u s ge p a c k t', 'un d', 'ge ö f f n e t .']

**Language Mitsmatch German training – English unknown text:**

segmented: ['T h e', 'c h i l d r en', 'a r e', 'p l a y i n g', 'w it h', 't h e i r', 'd o g s .', 'S h e', 'q u i c kl y', 'o p en e d', 't h e', 'b o x e s .', 'T h e', 'un h a p p y', 'st u d en t', 'd i s a g r e e d', 'w it h', 't h e', 't e a c h er .', 'T h e y', 'a r e', 'w a l k i n g', 'a n d', 's i n g i n g', 'i n', 't h e', 'g a r d en .', 'H e', 'm i s un d er st o o d', 't h e', 'q u e st i o n', 'a n d', 'r e p l a y e d', 't h e', 'v i d e o .']

<u>Overall interpretation Task 2:</u>

1. <u>Cross linguistic mismatch:</u>

- If you apply a German BPE model to English sentences (or vice versa), words are split into unnaturally tiny fragments or individual characters.
- The model cannot recognize typical prefixes, suffixes, or root words of the other language because its learned rules (merges) are based on the wrong statistical patterns.
- A sentence requires significantly more tokens in a mismatch scenario, which reduces processing efficiency.

2. <u>Rule-Based Stemming</u>

Strengths:

- It seems easy to emplement and relativly quick to execute. The computational costs seem relativly low.
- Cuts words down to stems. Good for Information Retrieval

Weaknesses

- Creates unnatural word forms and stems without linguistic meaning
- Cannot handle foreign words or grammar exceptions outside its predefined rules

3. Byte Pair Encoding (BPE):
   Strengths:

- can technically process any unknown word by falling back to smaller subwords or characters – (not considering the quality)
- adapts to frequency patterns without needing manual grammar rules

Weaknesses:

- Quality relies on the size and language of the training data
- performs poorly when applied to unseen languages